



Lesson 1.10 — Unit 1 Review and Assessment

Big idea: Unit 1 wraps up sequences and the foundations of functions — notation, domain/range, transformations, inverses, composition, and piecewise.

Quick review

Sequences: arithmetic adds d , geometric multiplies by r . Know both recursive and explicit forms.

Function notation: $f(\text{input}) = \text{output}$. $f(x + 2)$ means substitute the entire $(x + 2)$ for x .

Domain/range: avoid divide-by-zero, even roots of negatives, logs of non-positives.

Transformations: outside affects y the way you expect; inside affects x backwards.

Inverse / composition: inverses undo each other. Composition is order-sensitive: $f(g(x)) \neq g(f(x))$ in general.

Worked example

Worked review. Let $f(x) = 2x - 1$ and $g(x) = x^2$.

$(f \circ g)(3) = f(9) = 17$. $f^{-1}(x) = (x + 1)/2$. The graph of $g(x - 4) + 1$ is a parabola shifted right 4, up 1.

Warm-ups

1. Find a_{10} for the arithmetic sequence with $a_1 = 7$, $d = 4$.
2. Let $f(x) = 3x + 2$. Find $f(-5)$.
3. Find the domain of $f(x) = \sqrt{x - 1}$.

Practice

1. Find the inverse of $f(x) = 5x + 3$.
2. Let $f(x) = x^2$ and $g(x) = x - 4$. Find $f(g(x))$ and $g(f(x))$.
3. Describe the transformation that maps $y = |x|$ to $y = -2|x - 3| + 5$.

Challenge

1. A geometric sequence has $a_3 = 12$ and $a_6 = 96$. Find a_1 , r , and a_{10} .
2. Let $f(x) = \{ 2x + 1, x < 1 ; x^2, x \geq 1 \}$. Sketch (describe) the graph, evaluate $f(0)$, $f(1)$, $f(3)$, and decide whether f is continuous at $x = 1$.



Answer key — Lesson 1.10

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $a_{10} = 7 + 9(4) = 43$.
2. $f(-5) = -13$.
3. $x \geq 1$, i.e. $[1, \infty)$.

Practice

1. $f^{-1}(x) = (x - 3)/5$.
2. $f(g(x)) = (x - 4)^2$. $g(f(x)) = x^2 - 4$.
3. Shift right 3, vertical stretch by 2, reflect over x-axis, shift up 5.

Challenge

1. $r^3 = 96/12 = 8 \Rightarrow r = 2$. $a_1 = 12 / r^2 = 3$. $a_{10} = 3 \cdot 2^9 = 1,536$.
2. $f(0) = 1$, $f(1) = 1$, $f(3) = 9$. Top piece at $x = 1$ approaches 3; bottom piece equals 1. Different values $\rightarrow$ NOT continuous at $x = 1$ (a jump).